

HOMELAB DOCUMENTATION

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Infrastructure • Networking • Security

1. Project Overview

This homelab is built on a repurposed old PC running Proxmox VE as the bare-metal hypervisor. The primary goals are learning DevOps and networking practices while self-hosting personal applications in a controlled, production-like environment.

1.1 Goals

- Learn DevOps and networking hands-on in a real environment
- Self-host personal applications without depending on third-party cloud services
- Practice infrastructure concepts: containers, tunnels, TLS, access control
- Expose services securely to the internet without opening router ports

1.2 High-Level Architecture

Browser → Cloudflare Edge → [Zero Trust Auth] → Cloudflare Tunnel → cloudflared (LXC) → Proxmox (HTTPS :8006)

All external traffic enters through Cloudflare's global network. The cloudflared daemon running inside an LXC container maintains a persistent outbound tunnel — no inbound ports are ever opened on the router.

2. Infrastructure

2.1 Hardware & Hypervisor

Component	Details
Hardware	Old PC (repurposed)
Hypervisor	Proxmox VE
Proxmox Web UI	https://192.168.1.2:8006
Remote Access URL	https://proxmox.homelabpaulobarros.com
TLS Certificate	Self-signed (issued for 192.168.100.2)

2.2 Network Layout

Host / Service	IP Address	Role
Proxmox	192.168.1.2	Hypervisor — HTTPS on :8006
cloudflared LXC	192.168.1.82	Tunnel daemon container
Subnet	192.168.1.x	Primary LAN

2.3 LXC Containers

Containers run as unprivileged LXC instances on Proxmox. This keeps services isolated from the host and from each other.

- cloudflared runs in a dedicated Debian 12 LXC at 192.168.1.82
- Additional services can be added as separate LXC containers routed through the same tunnel

3. Cloudflare Tunnel

3.1 How the Tunnel Works

The cloudflared daemon initiates a persistent outbound connection to Cloudflare using QUIC (HTTP/2 as fallback). When a user hits proxmox.homelabpaulobarros.com, Cloudflare routes the request down this tunnel to cloudflared, which forwards it to Proxmox. The home network never receives unsolicited inbound connections.

3.2 Tunnel Details

Property	Value
Tunnel ID	bdf594e6-078f-4311-9bc4-c53222ef2710
Tunnel name	Proxmox
cloudflared version	v2026.3.0
Install mode	Config-file mode (systemd)
Config path	/etc/cloudflared/config.yml
Credentials path	Defined in config.yml
Origin service	https://192.168.1.2:8006

3.3 config.yml

The systemd unit references the config file directly (not a token), so all ingress rules in config.yml are respected:

```
tunnel: bdf594e6-078f-4311-9bc4-c53222ef2710
credentials-file: /etc/cloudflared/<tunnel-id>.json

ingress:
  - hostname: proxmox.homelabpaulobarros.com
    service: https://192.168.1.2:8006
    originRequest:
      noTLSVerify: true
  - service: http_status:404
```

3.4 Systemd Service

The service is configured in config-file mode (not token mode). Token mode ignores local config files entirely.

```
ExecStart=/usr/bin/cloudflared --config /etc/cloudflared/config.yml tunnel run  
Proxmox
```

Key lesson: Token-based installs (--token flag) ignore /etc/cloudflared/config.yml. Always verify which mode is active before debugging config changes.

3.5 TLS Handling

Proxmox uses a self-signed certificate issued for 192.168.100.2, while cloudflared connects to 192.168.1.2. The noTLSVerify: true ingress option bypasses certificate validation for the internal hop. TLS between the browser and Cloudflare's edge remains fully valid.

Future improvement: Issue a Let's Encrypt certificate for Proxmox via Proxmox's built-in ACME/DNS challenge using a Cloudflare API token. This would eliminate the need for noTLSVerify entirely.

4. Cloudflare Zero Trust Access

4.1 Overview

Cloudflare Access sits in front of the tunnel at Cloudflare's edge. Unauthenticated requests are intercepted and challenged before they are ever forwarded to cloudflared or Proxmox. This provides a second authentication layer independent of Proxmox's own login.

4.2 Access Policy Configuration

Setting	Value
Application type	Self-hosted
Application name	Proxmox
Subdomain	proxmox
Domain	homelabpaulobarros.com
Policy name	Allow me
Action	Allow
Include rule — Selector	Emails
Include rule — Value	(your email address)
Session duration	24 hours (recommended)

4.3 Authentication Flow

- User visits proxmox.homelabpaulobarros.com
- Cloudflare intercepts the request and shows a login page
- User authenticates via email OTP
- On success, Cloudflare issues a JWT and forwards the request through the tunnel
- Proxmox sees an already-authenticated request

4.4 Notes

- No cloudflared config changes are needed — Access is configured entirely in the Cloudflare Zero Trust dashboard
- The free Zero Trust plan covers up to 50 users; email OTP costs nothing
- Cloudflare Access does not replace Proxmox's own authentication — both layers are active

5. Key Learnings & Troubleshooting Notes

5.1 Token Mode vs Config-File Mode

The most impactful discovery during setup: when cloudflared is installed via a tunnel token (`--token` flag), the systemd unit ignores `/etc/cloudflared/config.yml` entirely. Any changes to `noTLSVerify` or ingress rules in the config file will have no effect until the service is switched to config-file mode.

Mode	How to identify	Config file respected?
Token mode	ExecStart contains <code>--token</code> flag	No
Config-file mode	ExecStart uses <code>--config</code> flag	Yes

5.2 CLI Flag Ordering

When using the CLI with `--no-tls-verify`, the flag must come before the `run` subcommand:

```
# Correct
cloudflared tunnel --no-tls-verify run --token <token>

# Wrong - flag is ignored
cloudflared tunnel run --no-tls-verify --token <token>
```

5.3 Origin URL Must Use HTTPS

Proxmox only listens on HTTPS port 8006. Setting the origin service to `http://` will fail. Always use `https://192.168.1.2:8006` as the origin.

5.4 Security Reminders

- Tunnel tokens are sensitive credentials — never share them in chat or commit them to version control
- If a token is accidentally exposed, revoke it immediately via the Cloudflare Zero Trust dashboard
- The Cloudflare dashboard UI does not expose a 'No TLS Verify' toggle for public hostnames — it must be set in `config.yml`

6. Roadmap

Item	Status	Notes
Proxmox on Cloudflare Tunnel	Complete	Config-file mode with noTLSVerify
Cloudflare Zero Trust Access	Complete	Email OTP policy in front of Proxmox
Let's Encrypt cert for Proxmox	Planned	Via ACME/DNS challenge + Cloudflare API token
Additional services via tunnel	Planned	Multiple subdomains on same tunnel
System design interview lab	Exploring	Side project, scope TBD

7. Quick Reference

7.1 Useful Commands

Check cloudflared service status:

```
systemctl status cloudflared
```

Restart cloudflared:

```
systemctl restart cloudflared
```

View live logs:

```
journalctl -u cloudflared -f
```

List tunnels:

```
cloudflared tunnel list
```

Test tunnel connectivity:

```
cloudflared tunnel info bdf594e6-078f-4311-9bc4-c53222ef2710
```

7.2 Important URLs

Resource	URL
Proxmox Web UI (local)	https://192.168.1.2:8006
Proxmox (remote)	https://proxmox.homelabpaulobarros.com
Cloudflare Zero Trust	https://dash.cloudflare.com → Zero Trust